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Wild ActionFormer: Enhancing wildlife action recognition for 11 endangered species in Wolong

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ABSTRACT

The video of wild animals captured by trap cameras provides conservationists with intuitive information on animal action, holding significant potential in ethology and ecology. This study focuses on 11 endangered wild animal species videos in the Wolong Nature Reserve and develops a video self-supervised learning-based animal action recognition network—Wild ActionFormer, to achieve automated analysis of wild animal action classes. We utilize UniformerV2 as the base backbone network, integrating self-supervised learning methods to enhance feature extraction capabilities. We constructed a differential dispersion regularization loss function to maintain the alignment of self-supervised learning features and improve the network's robustness against interference. The introduction of the Focal Loss reweighting strategy optimizes the loss for long-tail classes, mitigating the bias towards head data. Experimental results on our released LoTE-Animal open-source dataset show that the proposed action recognition network achieves a Top-1 accuracy of 95.09 , approximately 4 percentage points higher than the baseline. The LoTE-Animal dataset comprises 10k videos, including endangered wild animals from the Wolong Nature Reserve in Sichuan, China, such as the giant panda, sambar, and Sichuan snub-nosed monkey.

1 Introduction

Biodiversity is declining at an alarming rate. Wildlife populations have dropped significantly over the past few decades. According to the World Wildlife Fund (WWF), global wildlife numbers have fallen by 69 since 1970 (Almond et al., 2022). The International Union for Conservation of Nature (IUCN) reports that about 28 of species are now at risk of extinction (International Union for Conservation of Nature, 2024). The main causes include habitat destruction, climate change, poaching, and human activities. To combat this crisis, countries and organizations have taken action. Major agreements like the Convention on Biological Diversity (CBD) and CITES aim to protect species worldwide (Prip, 2024). Governments have also established nature reserves, ecological corridors, and breeding programs (Hilty et al., 2020). While these efforts have slowed biodiversity loss, effective wildlife monitoring remains a challenge. Nature reserves are crucial for conservation. They provide safe habitats and help species recover (Lopoukhine et al., 2012). Research shows that in protected areas, wildlife populations often rebound. For instance, the giant panda was reclassified from endangered to vulnerable in 2016 (Wei et al., 2018). However, reserves also face challenges. Many struggle with limited funding and staff, making long-term monitoring difficult. The Wolong Nature Reserve in China is a key sanctuary for endangered species like giant pandas and golden snub-nosed monkeys. Conservation efforts there include habitat expansion, stricter patrols, and breeding centers (Wenshi, 2014). However, monitoring animal behavior is still difficult. The rugged terrain makes it hard to set up reliable communication networks. Researchers

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must physically retrieve camera data and replace batteries (Wang et al., 2024). Later, they must manually review hours of footage. This process is slow and labor-intensive, delaying research (Liu et al., 2023b). The lack of automated behavior recognition is a major barrier to improving conservation efforts.

Deep learning has become a promising tool for automated wildlife action recognition. Researchers have applied these methods to species such as Amur tigers (Varun and Nagarajan, 2023), rhesus monkeys (Yang et al., 2023), and antelopes (Gübert et al., 2022). Schindler and Steinhage (2021) used deep learning to identify deer, boars, foxes, and hares in wildlife videos. They tested four CNN-based action recognition networks to analyze animal behavior. Similarly, Velasco-Montero et al. (2024) integrated AI algorithms into intelligent monitoring systems. Their approach enabled off-site processing and automated behavior analysis. Tripathi et al. (2025) used a UAV-equipped YOLO to detect and count 77 globally threatened swamp deer (*Rucervus duvaucelii*) in real time. Self-supervised learning (SSL) has also shown great potential for ecological monitoring. This method learns from data structure without relying on manual labels. In wildlife research, SSL helps models pre-train on large video datasets. This improves feature extraction and reduces the need for human annotation (Bi et al., 2023). Contrastive learning is one effective SSL technique. It creates positive and negative sample pairs from consecutive video frames. This helps models distinguish species and behavior patterns (Jia et al., 2022). Another approach, temporal consistency learning, improves recognition accuracy. It trains models to understand movement dynamics over time (Ilin et al., 2025). These advancements offer valuable insights for wildlife conservation and management. They improve monitoring efficiency and help researchers study animal behavior more effectively.

Animal behavior in the wild is influenced by circadian rhythms, seasonal changes, and climate conditions (Helm et al., 2013). A reliable model must perform well across these diverse ecological contexts. Wildlife datasets also tend to have long-tailed distributions, where some behaviors appear far more frequently than others (Liu et al., 2023a; Yu et al., 2021). To address these challenges, we used Uniformer (Li et al., 2023) as the backbone network. We also integrated a feature adapter to improve feature extraction. Since the adapter lacks self-supervised learning capabilities, we designed a custom loss function. This function improves feature alignment between the encoder and decoder, enhancing model robustness. Additionally, we applied reweighting techniques to handle the long-tailed distribution and reduce bias toward frequent behaviors.

We trained our model on the LoTE-Animal dataset (Liu et al., 2023a), which closely matches the Wolong Reserve's ecological conditions. This dataset contains 10,000 wildlife video samples. As shown in Fig. 1, our Wild ActionFormer model achieves faster training speeds and higher accuracy. It correctly identifies 21 typical behaviors across 11 Artiodactyla species, with a Top-1 accuracy of 95.09 .

2 Preliminary work

2.1. Action recognition

Action recognition models fall into two main categories: CNN-based and Vision Transformer (ViT)-based models (Dosovitskiy et al., 2020). CNN-based models include early designs like C3D (Tran et al., 2015), TSN (Wang et al., 2016), I3D (Carreira and Zisserman, 2017), and R(2+1)D (Tran et al., 2018). Later models, such as TSM (Lin et al., 2019), SlowFast (Feichtenhofer et al., 2019), TPN (Yang et al., 2020), and TANet (Liu et al., 2021a), further improved performance. CNNs excel at extracting spatial features from images but struggle to capture temporal dependencies in video sequences. ViT-based models – such as TimeSFormer (Bertasius et al., 2021) and Video Swin Transformer (Liu et al., 2022) – use self-attention mechanisms to capture long-range dependencies. They outperform CNNs in processing temporal information but often involve redundant global computations. Each approach has strengths and weaknesses. CNNs focus on spatial feature extraction, while ViTs handle temporal dependencies more effectively. To combine the benefits of both, Li et al. (2023) proposed Uniformer. This model integrates CNN-style hierarchical structures with Transformer-style modules, making it one of the best-performing backbone networks. However, Uniformer has limitations. It requires large-scale datasets for pre-training and fine-tuning. This makes it difficult to apply to tasks with limited data.

2.2. Video self-supervised learning

Self-supervised learning learns from data without labeled examples. It uses proxy tasks to extract meaningful features (Liu et al., 2021b). Methods like SimCLR (Chen et al., 2020) and MoCo (He et al., 2020) improve image classification and object detection by learning from unlabeled data. They compare different learning strategies to develop efficient feature representations. In speech processing, models like HuBERT (Hsu et al., 2021) and Wav2Vec 2.0 (Baevski et al., 2020) predict masked audio clips. This improves automatic speech recognition. However, these methods need large datasets and high computing power, leading to high application costs. Autoencoders are a common self-supervised model. They use an encoder to compress data and a decoder to reconstruct it. This process helps models self-learn and reconstruct images (Hinton and Salakhutdinov, 2006). For videos, similar techniques capture both spatial and temporal representations, as seen in Video MAE (Tong et al., 2022). To measure reconstruction accuracy, researchers often use Mean Squared Error (MSE). It calculates the average squared difference between original and reconstructed frames. MSE loss converges quickly but is sensitive to outliers, which can cause overfitting (Sai et al., 2009). Regularization techniques help prevent overfitting. Adding L1 or L2 penalties limits model complexity and improves generalization (Santos and Papa, 2022; Ng, 2004). The regularization coefficient plays a key role. It balances reconstruction loss and penalty terms, ensuring better model performance.

2.3. Long-tail data

Long-tail data creates class imbalance in visual recognition. Deep learning models often favor dominant classes, reducing accuracy for rare classes. Two common solutions are resampling and loss reweighting (Zhang et al., 2021). Resampling adjusts how often different classes appear in training. It includes undersampling dominant classes (Chawla et al., 2002; Han et al., 2005), oversampling rare classes (Drummond and Holte, 2003), and class-balanced sampling (Shen et al., 2016; Mahajan et al., 2018). These methods use inverse weighting to balance class distributions (Ding et al., 2020). However, repeated sampling of rare classes can reduce model robustness by overfitting to limited examples. Loss reweighting takes a different approach. It modifies the loss function to give higher weights to rare classes and lower weights to frequent ones (Li et al., 2020; Tan et al., 2021). Standard cross-entropy loss (Bishop, 1995) does not adjust for imbalances. Adding class weights helps correct this by penalizing errors on rare classes more heavily (Phan and Yamamoto, 2020).

The following sections build on these techniques. We present our approach, which addresses model limitations, especially in low-data scenarios and imbalanced datasets.

3 Materials and methods

3.1. Materials

Wolong National Nature Reserve (hereafter the reserve) is located between 102°52′ to 103°24′ E and 30° 45′ to 31°25′ N. It covers 2,000 km^2 and has an elevation range of 1,150 to 6,250 meters (Liu et al., 2001).

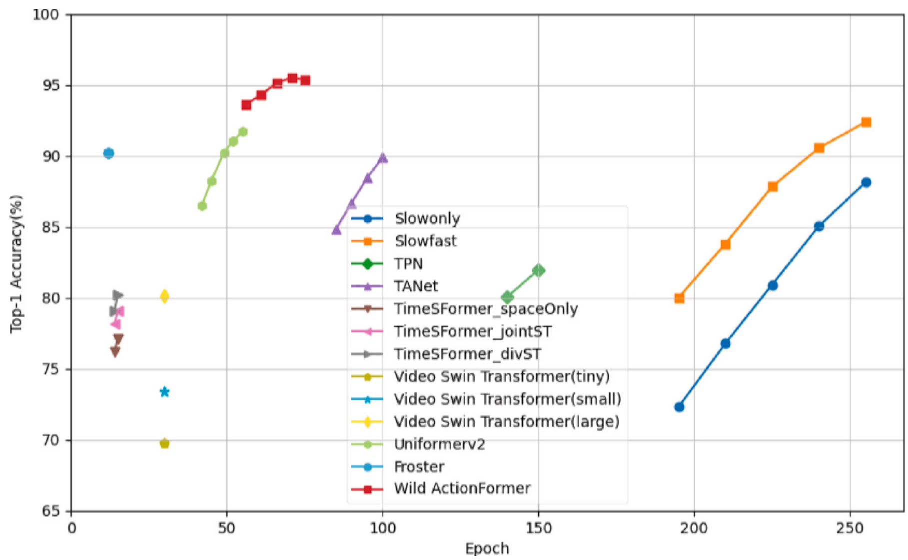

Fig 1 Comparison of mainstream methods on the video LoTE-Animal dataset.

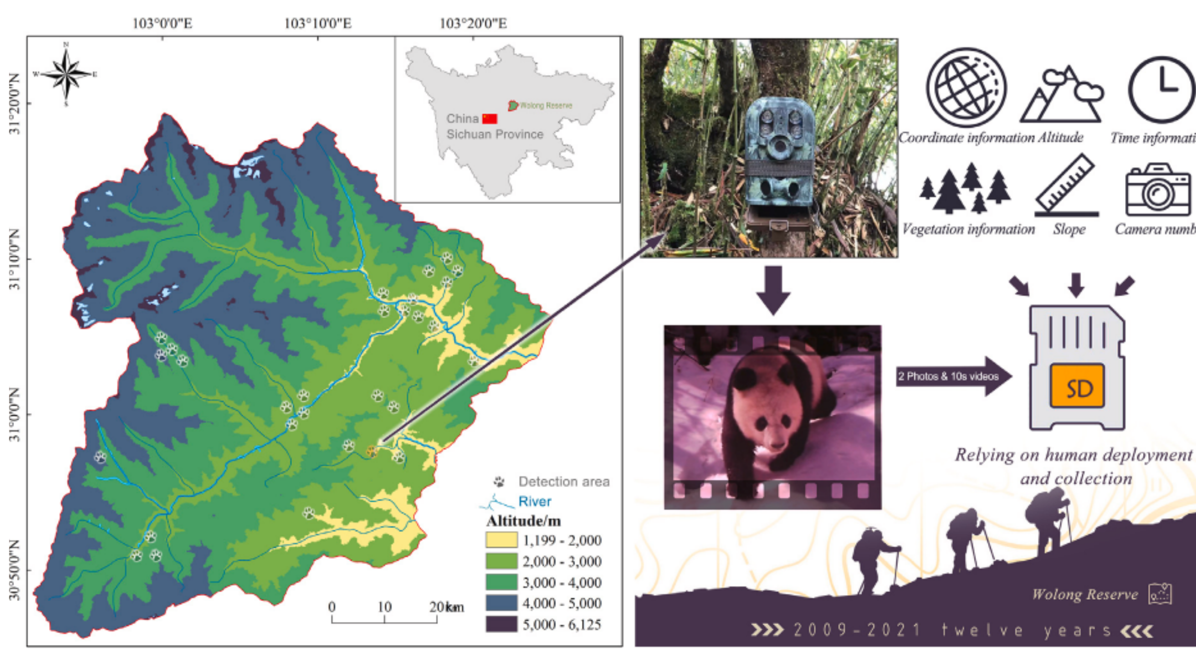

Fig 2 Monitoring sites in Wolong Nature Reserve.

3.1.1. Dataset construction

As shown in Fig. 2, infrared trap cameras (Ltl 5210 A) were mounted on tree trunks or other fixed objects. They were positioned to monitor passing wildlife. Each camera operated continuously. Upon triggering, it captured two images and a 10-second video. The recorded metadata included coordinates, altitude, slope, aspect, installation timestamp, camera ID, and vegetation type. Cameras were checked monthly or bimonthly, with battery and SD card replacements as needed. Camera placement was strategic. Locations were chosen based on feces, footprints, and feeding traces.

As the number of cameras increased each year, the amount of video data grew rapidly. Manual processing became more time-consuming. This challenge is common in other reserves, where managing large datasets requires efficient analysis systems.

3.1.2. Dataset content

We used the LoTE-Animal dataset, an infrared camera trap dataset that includes endangered species from the reserve. This dataset helps researchers study wildlife behavior. The study area contains diverse habitats with various animal species. Their activity patterns change with seasons, weather, and time of day, as shown in Fig. 3.

The dataset categorizes species across three dimensions. Scene Dimension – It includes time of day (morning, noon, night), seasonal changes (spring, summer-autumn, winter), and weather conditions (foggy, snowy, sunny, rainy, overcast, cloudy). These factors influence wildlife activity. Place Dimension – It covers different habitats, such as ponds, trees, ridges, rocks, seam nests, animal paths, ice surfaces, and cloud seas. These locations reflect the spatial diversity of wildlife behavior. Action Dimension – It describes animal behaviors, including

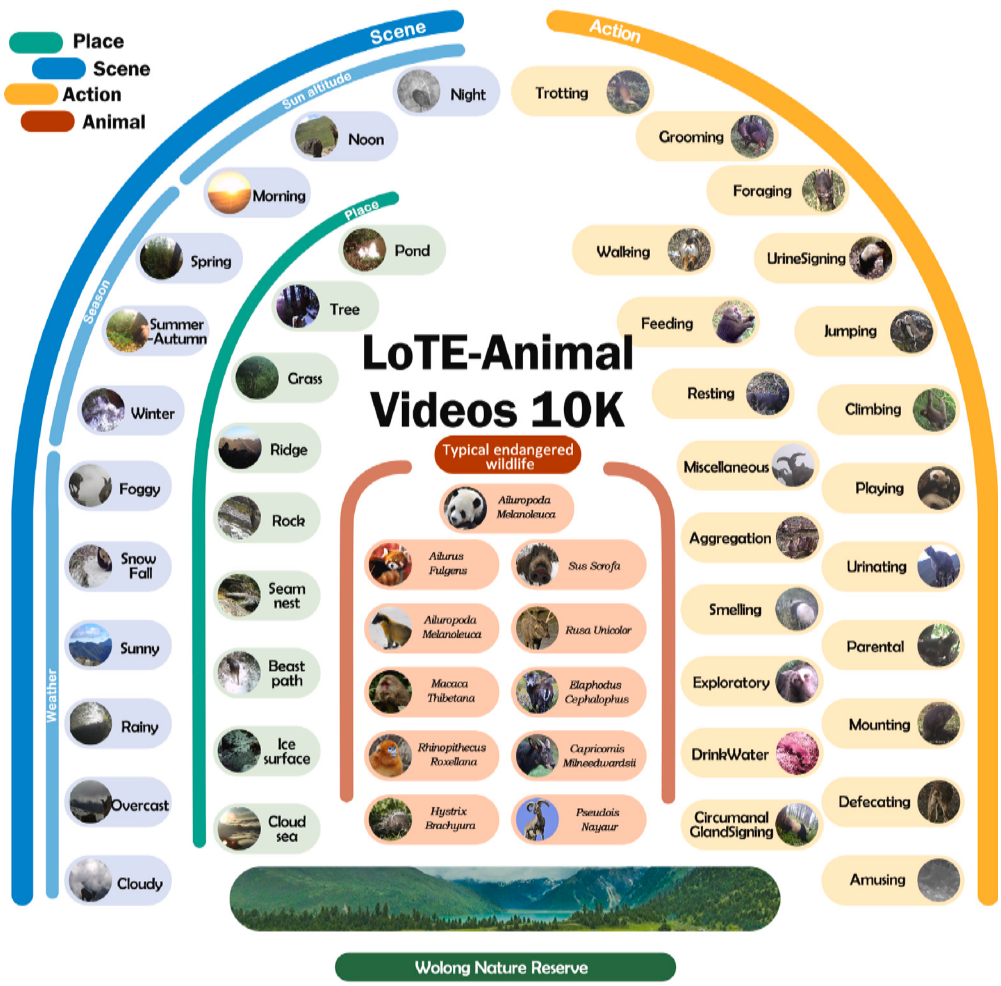

Fig 3 Construction of wild animal typical action dataset. Animals listed in Appendix A, with two species under Class I national protection, eight under Class II, and one under Class III.

locomotion (walking, trotting, jumping, climbing), social interactions (aggregation, playing, mounting, parental care), and physiological activities (urination, defecation, drinking, circum-anal gland marking). This structured classification provides a comprehensive view of wildlife activity in the reserve.

3.2. Methods

The network architecture in Fig. 4 integrates a self-supervised learning adapter to improve wild animal action recognition (Section 3.2.1). A loss function combines differential divergence regularization with MSE to enhance feature alignment and robustness (Section 3.2.2). A loss re-weighting function reduces bias from the long-tailed data distribution (Section 3.2.3).

The model has four main components: input, feature adapter, backbone, and output. The input consists of infrared trap camera videos. The output provides predicted animal actions. The Feature Adapter is designed for self-supervised spatio-temporal learning. It follows an encoder–decoder structure. During pre-training, the model learns behavior features from unlabeled video data by performing future frame prediction and occlusion recovery. Inception modules with multi-scale convolutions (1 × 1, 3 × 3, 5 × 5) refine features and capture temporal dependencies. The decoder uses deconvolution (unConv2d) to reconstruct occluded features and reduce information loss caused by occlusions and noise. The backbone UniFormerV2 combines CNNs and Transformers to model short- and long-term dependencies. The Local UniBlock captures short-term patterns, while Global Cross MHRA aggregates cross-frame information for long-range dependencies. A multi-stage fusion mechanism integrates local and global features. Detailed explanations of key components are provided in Supplementary Material, Appendix C.

3.2.1. Feature adapter

Inspired by Gao et al. (2022), we integrated an encoder–decoder adapter before the backbone. This module consists of an encoder, a temporal feature learner, and a decoder. The encoder extracts spatial features, the temporal feature learner models temporal dynamics, and the decoder fuses these spatiotemporal representations to predict future frames.

Encoder. The encoder is composed of a series of four ConvNorm-ReLU units (Conv2d + LayerNorm + LeakyReLU) to distill spatial attributes from C channels across dimensions (H, W). The encoding sequence is:

$$x_i = \sigma\ \mathrm{GroupNorm}\ \mathrm{Conv2d}\ x_{i-1}))) \quad 1 \quad i \quad 4 \tag{1}$$

where x_{i-1} and x_i have dimensions $T\ C\ H\ W)$ and $T\ C'\ H'\ W')$, respectively.

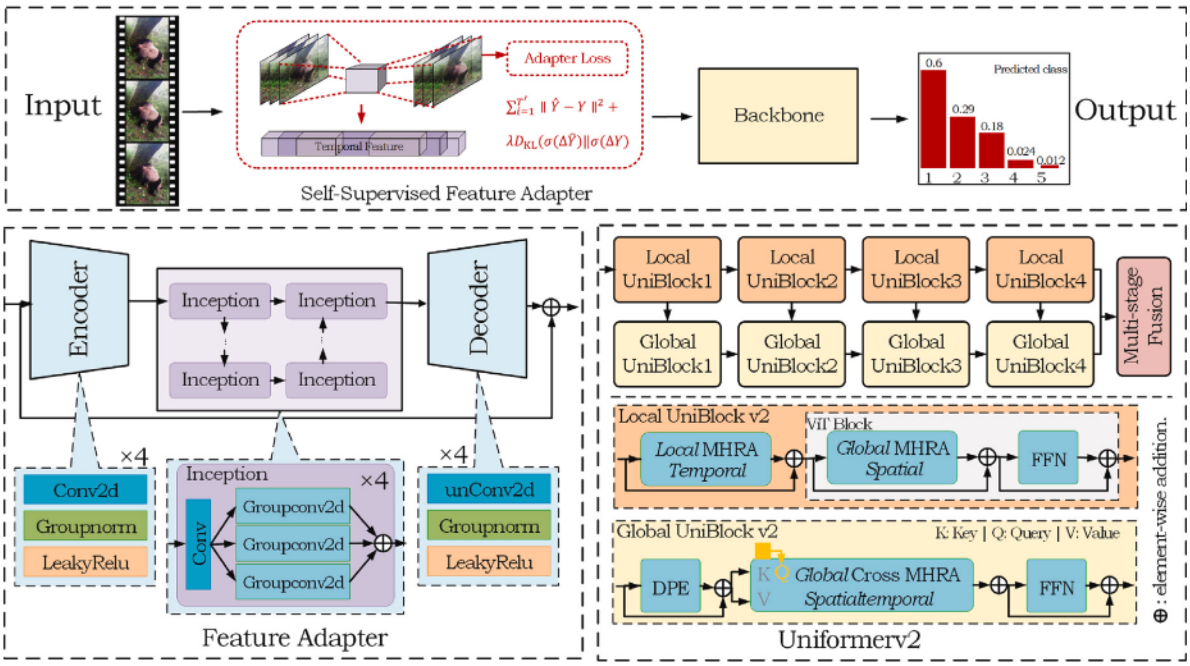

Fig 4 Overall network architecture.

Temporal Feature Learner. This module captures temporal evolution using Inception layers. It processes $T \times C$ channels over $H\ W)$. Each Inception module combines a bottleneck Conv2d (1×1 kernel) with parallel GroupConv2d operations. The stacking sequence for this layer is:

$$x_i = \sum \left(\text{Conv2d Conv } x_{i-1})_{1\times 1})\right)_{j\times j} \quad 5 \quad i \quad 8 \quad j = 3\ 5\ 7\ 11 \tag{2}$$

where x_{i-1} and x_i have dimensions $T \times C\ H\ W)$ and $T'\ C'\ H'\ W')$, respectively. Decoder. The decoder reconstructs actual frames using four unConvNormReLU units (ConvTranspose2d + GroupNorm + LeakyReLU). It processes C' channels over $H'\ W')$. The decoding follows:

$$x_i = \sigma \text{ GroupNorm unConv2d } x_{i-1}))) \quad 9 \quad i \quad 12 \tag{3}$$

where x_{i-1} and x_i have dimensions $T'\ C'\ H'\ W')$ and $T\ C\ H\ W)$, respectively.

3.2.2. Differential divergence regularization

Unlike Gao et al. (2022), we go beyond integrating an adapter. We also introduce a custom loss function. Specifically, we apply differential divergence regularization on top of MSE loss. This ensures better alignment between generated representations and input representations. This loss function transforms differences between predicted and actual frames into a probability distribution (Tan et al., 2023). By computing the Kullback–Leibler (KL) divergence between predicted and actual frames, the model learns to capture patterns of change in the video.

For the predicted frame sequence $\hat{Y}$ and actual sequence Y, we first calculate the forward differences:

$$\Delta\hat{Y}_i = \hat{Y}_{i+1} - \hat{Y}_i \tag{4}$$

$$\Delta Y_i = Y_{i+1} - Y_i. \tag{5}$$

Next, we convert these differences into probabilistic values using the softmax function over the channel, height, and width dimensions to yield $\sigma\ \Delta\hat{Y})\ \sigma\ \Delta Y)$:

$$\sigma\ \Delta\hat{Y})_{i\,j\,k\,l} = \frac{\exp\ \Delta\hat{Y}_{i\,j\,k\,l}/\tau)}{\sum_{j'=1}^{C}\sum_{k'=1}^{H}\sum_{l'=1}^{W}\exp\ \Delta\hat{Y}_{i\,j'\,k'\,l'}/\tau)} \tag{6}$$

$$\sigma\ \Delta Y)_{i\,j\,k\,l} = \frac{\exp\ \Delta Y_{i\,j\,k\,l}/\tau)}{\sum_{j'=1}^{C}\sum_{k'=1}^{H}\sum_{l'=1}^{W}\exp\ \Delta Y_{i\,j'\,k'\,l'}/\tau)} \tag{7}$$

where τ is a temperature parameter, empirically set to 0.1, which refines the probability distributions. The softmax function emphasizes frames with significant differences, penalizing them more heavily. We define the sequential frame divergence regularization loss as the KL divergence between the probability distributions:

$$L_{\text{reg}}\ \hat{Y}\ Y) = D_{\text{KL}}\ \sigma\ \Delta\hat{Y})\ \|\ \sigma\ \Delta Y)) = \sum_{i=1}^{T'-1} \sigma\ \Delta\hat{Y}_i) \log \frac{\sigma\ \Delta\hat{Y}_i)}{\sigma\ \Delta Y_i)}. \tag{8}$$

Our model is trained end-to-end in two steps. The total loss function combines MSE loss and sequential frame divergence regularization, weighted by λ:

$$L = \sum_{i=1}^{T'}\ \hat{Y} - Y\ ^2 + \lambda L_{\text{reg}}\ \hat{Y}\ Y). \tag{9}$$

3.2.3. Loss re-weighting

Loss re-weighting improves model performance by adjusting penalization strategies (Fu et al., 2022). It tackles long-tailed distribution by modifying loss weights for head and tail classes (Li et al., 2020; Tan et al., 2021). Our dataset follows a long-tailed distribution (Fig. 5). The four head categories make up 72 , while the 17 tail categories account for 28 . This imbalance skews model learning.

Standard multi-class cross-entropy (CE) loss does not adjust for class imbalance (Bishop, 1995). Adding class weights helps balance sample importance (Phan and Yamamoto, 2020). Typically, minority classes receive higher weights, while majority classes receive lower weights. The weighted cross-entropy loss is defined as:

$$\alpha\text{-balanced CELoss} = -\sum_{i=1}^{N} \alpha_{y_i} \cdot \log\ p_{y_i}) \tag{10}$$

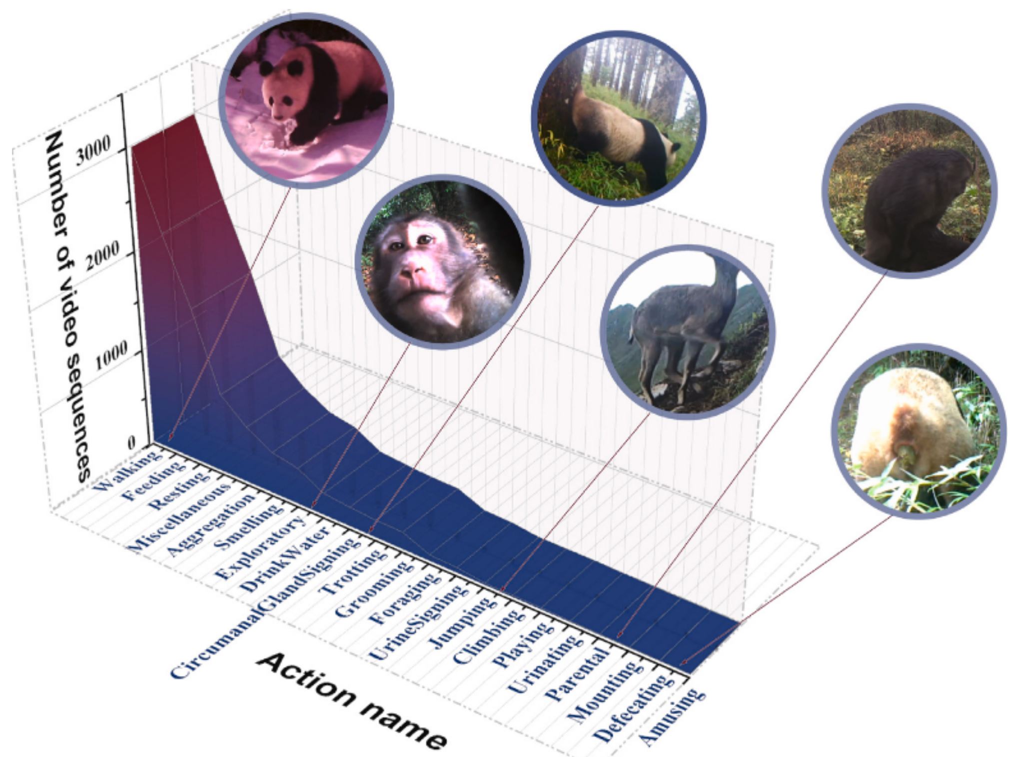

Fig 5 Distribution of wildlife action classes.

where N is the number of samples, p_{y_i} is the predicted probability for the actual class y_i, and α_{y_i} is the weight for actual class. Cross-entropy loss treats all samples equally. However, it does not distinguish between easy and hard samples. To address this, we use focal loss (Lin et al., 2017). Focal loss dynamically reduces the weight of well-classified samples during training. It introduces a modulation factor $1 - p_{y_i})^{\gamma}$, where $\gamma \geq 0$, The modified loss function is:

$$\text{Focal Loss } p_{y_i}) = -\alpha_{y_i} \; 1 - p_{y_i})^{\gamma} \log p_{y_i}) \tag{11}$$

where α_{y_i} adjusts for class imbalance, while γ controls the focus on hard samples. We set $\gamma = 2$, following best practices.

Focal loss helps by increasing the impact of hard-to-classify minority samples, making the model more robust in handling imbalanced data.

4 Results and discussion

This study was conducted on four 2080Ti GPUs, providing a total of 48 GB of memory. We used the default parameters from the MMLAB (MMAction2 Contributors, 2020) library for model optimization. The batch size for training, validation, and testing was set to 2. The initial learning rate was 1×10^{-5}, and we used the AdamW optimizer with betas set to (0.9, 0.999) and a weight decay of 0.05. The model was trained for up to 75 epochs. Data augmentation strategies included uniform sample for random sampling, rand augment for data augmentation, and random resized crop for cropping. We applied a 50 probability of using the flip operation. The learning rate schedule followed a two-phase strategy. In the first 5 epochs, we applied linear growth, increasing the learning rate from 0.1 times the initial value to the full initial value. After that, we used cosine annealing LR for the remaining epochs, gradually decreasing the learning rate to 10 of its original value. For optimization, we used both MSE Loss and differential divergence regularization loss. We also applied gradient clipping to stabilize training. The weight decay was carefully tuned, but bias and normalization layer parameters were excluded from weight decay. For preprocessing, the input format was set to (N, C, T, H, W). Images were cropped to 224 × 224 to match the model's input size and normalized using the mean and standard deviation of the pixel values. Data was decoded with decord. During training, we used random resized crop, while for testing and validation, three crop and center crop were used for standardization.

4.1. Comparison with previous work

We compared our model with previous works on the LoTE-animal dataset, evaluating each model's performance using Top-1, Top-5 accuracy, GFLOPs/view and params. The results are presented in Table 1.

Wild ActionFormer outperformed other models on the LoTE-animal dataset. It achieved 95.09 Top-1 accuracy and 99.9 Top-5 accuracy. In comparison, the swin-large variant of Video Swin Transformer had a slightly higher Top-5 accuracy of 99.95 , but its Top-1 accuracy was lower at 92.81 . This shows that our model is better at identifying the primary action class. When compared to models like SlowOnly and SlowFast, Wild ActionFormer achieved better Top-1 accuracy at the same sampling frame rate. This highlights the impact of architectural and algorithmic improvements on recognition performance. TimeSFormer, on the other hand, showed weaker performance, pointing to its limitations in capturing dynamic and complex spatiotemporal features.

Our comparison reveals that CNN-based models are efficient but struggle with complex temporal dependencies. ViT-based models are powerful but computationally expensive. Froster, a recent Transformer-based model, also shows competitive performance with a good balance between accuracy and efficiency. Wild ActionFormer and Uniformerv2, both hybrid CNN+ViT models, perform better than pure CNN models

Table 1
Quantitative comparison of different methods on the LoTE-Animal dataset. ⋆ denotes CNN, denotes ViTs, and • represents CNN+ViTs hybrid architecture, indicating different backbone.

Model	Year	Backbone	Frame sample	Top-1( )	Top-5( )	GELOPs	Params
SlowOnly	ICCV2019	ResNet50 ⋆	4 × 16 × 1	90.13	99.36	41.92	31.68M
SlowFast	ICCV2019	ResNet50 ⋆	4 × 16 × 1	92.96	99.4	6.96	33.61M
TPN	CVPR2020	ResNet50 ⋆	8 × 8 × 1	90.33	99.26	50.46	90.72M
TANet	ICCV2021	ResNet50 ⋆	1 × 1 × 8	91.32	99.36	32.95	24.81M
TimeSFormer	ICML2021	spaceOnly	8 × 32 × 1	74.06	98.46	141	85.82M
		jointST	8 × 32 × 1	78.37	98.56	18	85.82M
		divST	8 × 32 × 1	80.06	98.71	196	122M
Video Swin Transformer	CVPR2022	swin-tiny	8 × 32 × 1	84.18	99.55	19.77	28.16M
		swin-small	8 × 32 × 1	88.34	99.9	37.88	49.82M
		swin-large	8 × 32 × 1	92.81	99.95	144	197M
Uniformerv2	ICCV2023	Uniformerv2 •	8 × 8 × 1	91.17	99.85	143	128M
Froster	ICML 2024	Transformer	8 × 8 × 1	90.80	99.78	140	86M
Wild ActionFormer	Our 2025	Uniformerv2 •	8 × 8 × 1	**95 09**	**99 9**	**189**	**128M**

Table 2
Comparison of integrating the self-supervised feature adapter.

Self-supervised adapter	Training memory	Top-1( )	Top-5( )
	7817M	91.17	99.85
	6758M	92.81(+1.64)	99.75(−0.1)

Table 3
Comparison of self-supervised learning adapter loss function.

Self-supervised adapter loss	Top-1( )	Top-5( )
	92.81	99.75
MSE loss	93.01 (+0.2)	99.9 (+0.15)
Adapter loss	94.89 (+2.08)	99.8 (−0.05)

Table 4
Settings of the λ parameter in differential divergence regularization term.

λ Parameter	Top-1( )	Top-5( )
0.5	94.15	99.9
1	94.89	99.8
4	94.69	99.85
8	94.3	99.9

Table 5
Comparison of reweighted loss functions.

Reweighted loss functions	Top-1( )	Top-5( )	Head( )	Tail( )
CE loss	94.89	99.8	94.88	76.31
α-balanced CE loss	94.44	99.9	94.47	77.88
Label smoothing loss	95.04	99.9	95.01	80.81
Focal loss	95.09	99.9	94.56	82.71

by incorporating ViT components to improve temporal modeling. This leads to a significant boost in recognition performance. At the same time, they retain CNNs for local feature extraction, which reduces the reliance on computationally expensive self-attention layers found in pure ViTs.

4.2. Ablation studies

To evaluate our model's improvements, we conducted ablation experiments on three key components: the self-supervised feature adapter, differential divergence regularization loss, and loss re-weighting. These experiments assessed how each component contributed to wild animal action recognition accuracy.

4.2.1. Self-supervised adapter

To analyze the impact of the self-supervised adapter, we compared two versions of the model: one without the adapter and one with it. The results are shown in Table 2.

The results reveal that integrating the adapter increased Top-1 accuracy from 91.17 to 92.81 , a gain of 1.64 . However, Top-5 accuracy dropped slightly from 99.85 to 99.75 . This suggests that the adapter improved the model's ability to identify the primary action class but had a minor negative effect on recognizing the Top-5 action categories. Additionally, integrating the adapter reduced memory usage during training. This may be because the adapter helps the model learn more efficient feature representations, lowering resource demands.

4.2.2. Self-supervised adapter loss

We examined how the self-supervised adapter loss affects wild animal action recognition accuracy. The experiments were divided into three groups: one without a loss function, one using MSE loss, and one with the self-supervised adapter loss. The results are shown in Table 3.

Without a loss function, the model achieved 92.81 Top-1 accuracy and 99.75 Top-5 accuracy. Using MSE loss slightly improved performance, raising Top-1 accuracy to 93.01 and Top-5 accuracy to 99.9 . The self-supervised adapter loss further boosted Top-1 accuracy to 94.89 , though Top-5 accuracy dropped slightly to 99.8 . These results suggest that the self-supervised adapter loss enhances model performance, especially in identifying the primary action class.

We further explored the effect of differential divergence regularization parameter λ in the self-supervised adapter loss function. The results are shown in Table 4.

We found that different λ values caused only minor variations in performance. The best result was achieved with $\lambda = 1$, reaching 94.89 Top-1 accuracy. This indicates that λ is not highly sensitive in this setting, and $\lambda = 1$ was a suitable choice for our experiments.

4.2.3. Loss re-weighting

We compared four types of reweighted loss functions: standard cross-entropy (CE) loss, α-balanced CE loss, label smoothing loss, and focal loss. Table 5 shows the results.

The baseline model with CE loss performed poorly on tail categories, achieving only 76.31 accuracy. This was due to the limited number of samples in those categories, which led to insufficient learning. The α-balanced CE loss adjusted class weights to reduce the dominance of head categories. This slightly improved tail accuracy to 77.88 , while head category performance remained almost unchanged. Label smoothing helped reduce overfitting. It improved tail category accuracy to 80.81 and raised overall Top-1 accuracy to 95.04 . Focal loss placed more emphasis on hard-to-classify tail samples. As a result, tail accuracy improved significantly to 82.71 , with little effect on head category performance.

Overall, tail accuracy gradually increased across the four methods. This shows that reweighting strategies are effective in handling long-tail distributions. Focal loss achieved the best results by focusing training on challenging samples, especially in underrepresented

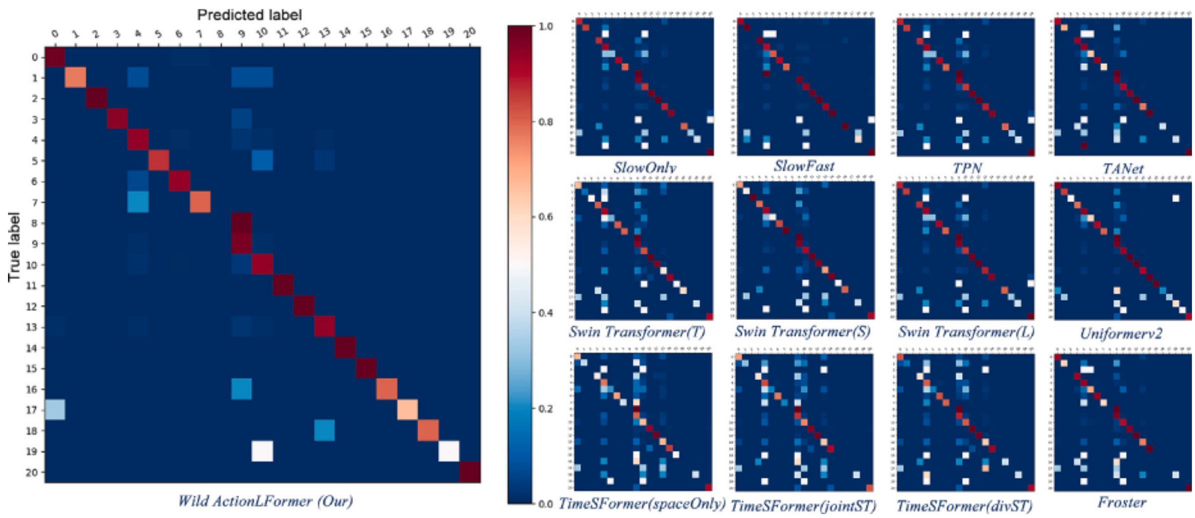

Fig 6 Comparison of confusion matrix for action recognition methods on the LoTE-Animal dataset. Swin Transformer (T/S/L) is Tiny/Small/Large.

classes. Importantly, both label smoothing and focal loss improved tail performance without compromising accuracy on head categories.

4.3. Qualitative analysis

4.3.1. Confusion matrix analysis

To evaluate model performance, we used confusion matrices to compare the classification accuracy of different video models on the animal action dataset. Fig. 6 shows the results. The horizontal axis represents predicted labels, while the vertical axis shows true labels. Color intensity indicates accuracy—deep red means high accuracy, and light blue means low accuracy.

The confusion matrix of Wild ActionFormer shows the best performance. Most of its values lie along the diagonal, indicating accurate predictions across most action classes. In contrast, other models such as SlowFast, TPN, and TA Net show more dispersed patterns. This suggests higher confusion and lower classification accuracy. Uniformerv2, Froster and Swin Transformer (T) perform relatively well, though they still misclassify some categories.

Some action classes stand out for their strong performance. Classes 9 (Walking) and 10 (Resting) show deep red diagonal blocks, meaning they are classified accurately. These actions have many training samples, helping the model learn their features effectively. Classes 12 (Drink Water) and 20 (Urine Signing) also show nearly perfect classification. Their distinct features likely make them easy for the model to identify. On the other hand, classes 2, 8, 15, and 19 (Mounting, Amusing, Defecating, Urinating) show weaker diagonal colors and more scattered predictions. These classes have fewer samples, making it harder for the model to learn meaningful features. Notably, class 8 (Amusing) is absent from the training set, so all models fail to recognize it. The serial numbers used in the confusion matrix correspond to the animal action classes listed in Appendix B.

4.3.2. Heatmap visualization

To further assess model performance, we used heatmap visualizations to analyze how different models recognize animal actions across video sequences. Fig. 7 shows the results. Each row represents a specific action. The columns display each model's key frame importance matrix and attention regions.

CNN-based models (SlowOnly, SlowFast, TPN) show narrow attention areas. These models often focus on specific local regions, which limits their ability to recognize complex actions. ViT-based models (TimeSFormer, Swin Transformer, Froster) show wider attention coverage. However, they often attend to irrelevant background areas, which can reduce accuracy. UniformerV2, which combines CNN and ViT structures, improves attention spread. Still, it struggles to focus on relevant parts for certain behaviors. In contrast, Wild ActionFormer shows precise and meaningful attention. It consistently focuses on key body parts that define the behavior. For example, in exploratory actions, it highlights the head and forelimbs. In feeding actions, it targets the mouth and food. During urine signing, it correctly identifies the hind limbs and interaction zones while ignoring background distractions. These results show that Wild ActionFormer makes better use of spatiotemporal features. Its attention maps align well with behavior semantics, improving recognition accuracy and interpretability.

4.3.3. Failure cases

To better understand the model's performance, we analyzed the test set and examined cases where the model made incorrect predictions, as shown in Fig. 8.

Wild ActionFormer performed well overall but faced challenges with fine-grained behavior classification. Most misclassifications occurred between visually similar actions, such as foraging vs. feeding and resting vs. walking. Transient behaviors like defecation and exploratory movement also caused confusion. Performance dropped noticeably on nighttime infrared images, likely due to reduced visibility. In some failure cases, the model gave high confidence to incorrect predictions. This suggests that its decision boundaries could be further refined. Future improvements may include better sensitivity to subtle motion cues, stronger use of temporal features, and enhanced feature extraction under low-light conditions.

4.4. Successful cases

Fig. 9 shows several challenging examples the model correctly classified.

In these cases, Wild ActionFormer accurately recognized behaviors like walking, exploration, and mounting with high confidence. It also maintained low probabilities for incorrect classes. These results show that the model is robust in complex environments and can distinguish key behaviors effectively. However, further refinement is still needed to handle fine-grained distinctions and difficult lighting conditions more reliably.

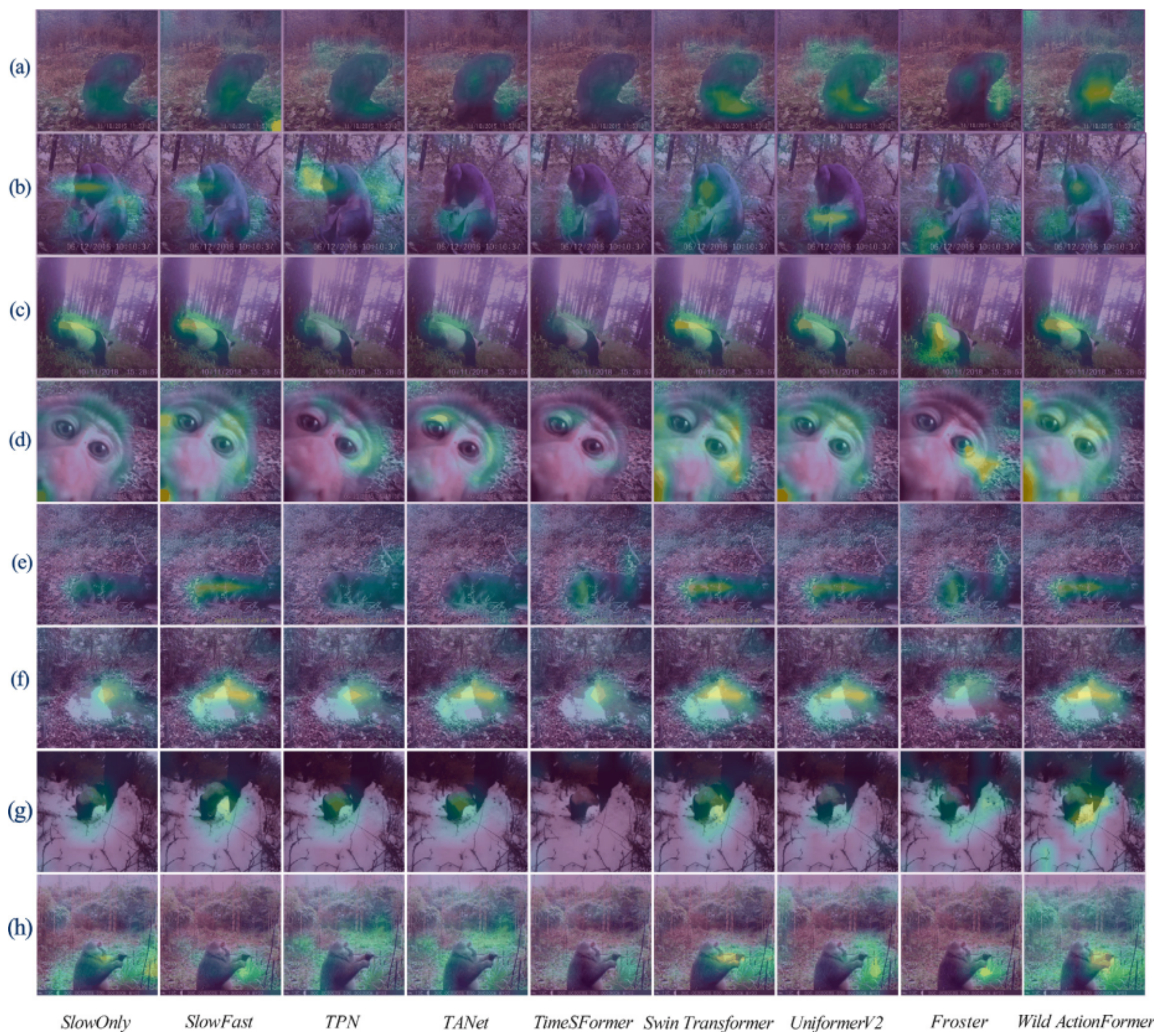

Fig 7 Visualization of importance weight matrix with Grad-CAM. (a) Mounting, (b) Grooming, (c) Circumanal Gland Signing, (d) Exploratory, (e) Resting, (f) Drink Water, (g) Urine Signing, (h) Feeding.

4.5. Analysis of seasonal ecological changes and action patterns

We analyzed animal behaviors predicted by Wild ActionFormer to explore how seasonal ecological changes affect action patterns in the Wolong Reserve. The reserve experiences three distinct seasons: spring, summer-autumn, and winter. These seasonal shifts significantly influence behaviors such as foraging, aggregation, and exploration. Understanding these changes helps reveal animals' adaptive strategies and survival mechanisms. To study the seasonal effects, we examined the monthly frequency and seasonal distribution of each action. Fig. 10 shows this data in separate subplots. Bubble size indicates action frequency. Colors represent seasons—spring (green), summer-autumn (orange), and winter (blue). The data spans January to December and is normalized from 0 to 10 for easier comparison.

The results show that aggregation behaviors peak in winter and summer-autumn. This likely reflects a need for warmth in winter and group gathering for food or reproduction during warmer months. Foraging is most common in spring, especially in June, when resources are abundant. Mounting increases in summer, aligning with breeding seasons. Trotting is frequent in summer and autumn, possibly related to foraging and regulating body temperature.

Feeding actions rise in July and September, again reflecting resource availability. Grooming is more frequent from spring to summer, likely linked to social bonding and coat care. Exploration increases during spring and summer, suggesting better environmental conditions and resource availability. Climbing peaks in August, October, and November, when conditions support wider movement. Walking occurs year-round but is most frequent in summer and autumn, when temperatures are moderate. Resting also increases in these seasons, likely to balance high activity levels. Drinking is more frequent in winter, possibly due to water scarcity. Circumanal gland marking, a behavior unique to pandas, appears often in winter and spring, indicating increased territorial activity during resource competition. Jumping is common in summer and autumn, possibly due to elevated energy and social play. Play behavior peaks in spring and early summer-autumn, especially among young animals in the breeding season. Urination and defecation increase in June and July, likely reflecting higher food intake and metabolism. Smelling and urine marking remain consistent year-round, indicating steady social and physiological needs.

When paired with seasonal analysis, Wild ActionFormer provides valuable insight into behavior changes related to breeding, migration, and foraging. This supports habitat conservation by identifying key

Fig 8 Failure examples from the test set.

periods and behaviors. For instance, protecting migration paths or ensuring undisturbed breeding zones. The model also helps track the impact of extreme weather and vegetation changes. For example, reduced activity after heatwaves or snowstorms could inform actions like adding water sources or limiting human activity in affected areas.

4.6. Analysis of diurnal cycles and action patterns

We analyzed Wild ActionFormer's predictions to explore how diurnal changes affect animal behavior in the Wolong Reserve. Animal activity is closely tied to the day-night cycle. Light levels and temperature shifts influence when and how animals behave. These patterns reflect both physiological needs and long-term adaptations. To understand this effect, we examined the hourly activity of different species. Fig. 11 shows these patterns across a 24-hour period. The x-axis represents time of day, while the y-axis shows activity level. Daytime is shown in light yellow, and nighttime in gray.

The results reveal three main activity patterns: diurnal, nocturnal, and bimodal. Diurnal species, such as the The giant panda (*Ailuropoda Melanoleuca*), red panda (*Ailurus Fulgens*), Tibetan macaque (*Macaca Thibetana*) and blue sheep (*Pseudois Nayaur*), are most active during the day. Their behaviors – mainly foraging and socializing – occur when light levels are high. In contrast, the Malayan Porcupine (*Hystrix Brachyura*) and Chinese Serow (*Capricornis Milneedwardsii*) are nocturnal. They are mainly active at night, likely to avoid predators and reduce competition. The golden snub-nosed monkey (*Rhinopithecus Roxellana*) and tufted deer (*Elaphodus Cephalophus*) show bimodal activity. They are active in the early morning and late afternoon, resting during midday and late night. These patterns highlight each species'

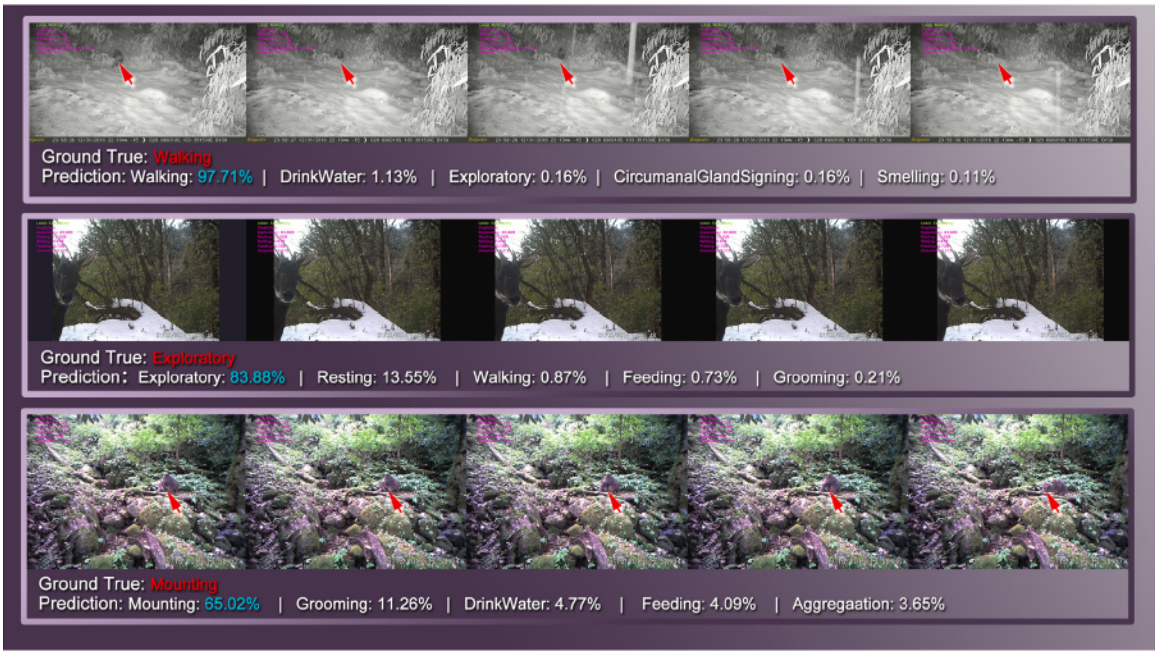

Fig 9 Successful examples from the test set.

adaptive strategies to light, temperature, and ecological pressures. Understanding these rhythms can guide conservation strategies. Ensuring access to resources at the right time of day supports animal health and survival.

Wild ActionFormer, when used with time-based analysis, can uncover these daily behavior patterns. It also helps identify human impact. For example, a shift toward nocturnal activity may suggest animals are avoiding daytime disturbances. This insight can help conservation managers adjust patrols or tourism schedules to reduce stress on wildlife.

5 Conclusion

In this study, we introduce Wild ActionFormer, an action recognition network designed specifically for wild animal behavior analysis in Wolong. The model uses UniformerV2 as its backbone and integrates self-supervised learning to improve feature extraction. To enhance alignment in the self-supervised process, we propose a differential divergence regularization loss. In addition, we apply a Focal loss reweighting strategy to address the challenge of long-tail class distributions. Wild ActionFormer achieves strong performance on wildlife action recognition tasks. Although Focal loss improves the classification of tail classes, some categories with very limited data still show poor performance. Future work could improve these results through data augmentation or by using cross-species data to enrich the tail classes. More flexible training strategies could also help the model better adapt to complex and dynamic environments. For real-world deployment, especially in resource-limited ecological reserves, we suggest using NVIDIA Jetson Xavier NX or Jetson Orin for high-precision, real-time monitoring where network infrastructure is available. In low-power scenarios, where real-time processing is less critical, Jetson Nano or Coral Edge TPU offer energy-efficient alternatives.

Looking forward, Wild ActionFormer could be integrated into a broader intelligent ecological monitoring system. For example, satellites and UAVs can track habitat changes, while IoT devices like infrared cameras and environmental sensors can support real-time behavior analysis. Edge computing devices can process data efficiently in the field, and UAVs can autonomously patrol remote areas, detect anomalies, and issue early warnings. In the cloud, AI-powered data fusion and forecasting can provide insights into the effects of climate change, habitat loss, and human activity. This system can offer strong scientific support for conservation planning and decision-making.

CRediT authorship contribution statement

Dan Liu: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Qianqian Sun:** Validation, Investigation, Data curation. **Jin Hou:** Validation, Investigation, Data curation. **Bochuan Zheng:** Resources, Project administration, Formal analysis. **Jindong Zhang:** Supervision, Resources, Investigation, Data curation. **Desheng Li:** Investigation, Data curation. **Tomas Norton:** Writing – review & editing, Supervision, Project administration. **Jifeng Ning:** Writing – original draft, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ecoinf.2025.103148.

Fig 10 Seasonal variation in animal action in Wolong Nature Reserve.

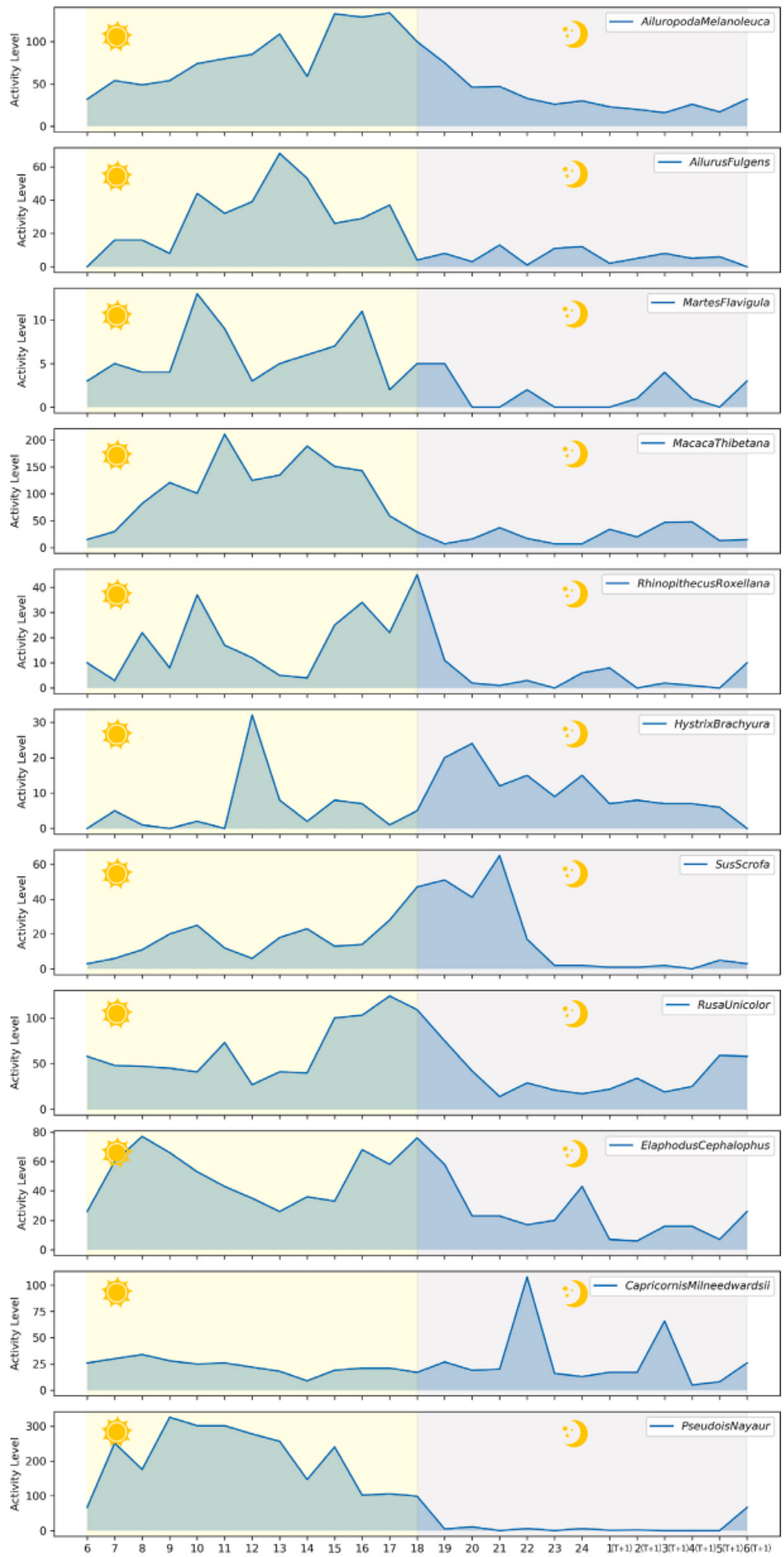

Fig 11 Diurnal variation in animal action in Wolong Nature Reserve.

Data availability

Dataset is available at https://lote-animal.github.io. Code is available at https://github.com/LoTE-Animal/Wild-ActionFormer.git.

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